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Astra Space

Software Engineer Position

Dear Hiring Manager,

I'm writing to express my strong interest in the Software Engineer position at Astra. As a full-stack engineer with a track record of building production systems under high-stakes conditions—from blockchain infrastructure to AI-powered enterprise tools—I'm excited by the opportunity to apply my skills to rocket testing, manufacturing, and launch operations.

What draws me to Astra is your data-driven approach to aerospace engineering and the emphasis on vertically integrated software. Throughout my career, I've thrived in environments where software directly impacts critical operations. At Centigrade, I rewrote API services in FastAPI and optimized database architectures that supported a 5x increase in query throughput while reducing latency by 60%. These optimizations required deep system design thinking and close collaboration with cross-functional teams—skills that translate directly to building reliable tools for rocket operations.

My experience aligns well with Astra's technical needs. I've built Python-based systems for real-time data processing, deployed cloud-native infrastructure on AWS and Kubernetes, and designed RESTful and microservice architectures that operate under complexity. At CarbonSustain, as founding engineer on a two-person team, I architected a blockchain-based tracking system from prototype to MVP, integrating real-time transaction validation and building CI/CD pipelines for zero-downtime releases. At Belden, I developed container ingress features and LLM-powered threat detection for industrial IoT systems, working directly with hardware teams to deliver software that had to work flawlessly in production environments.

I'm particularly excited about Astra's philosophy that sometimes the best rocket software requires the versatility of "programming a synthesizer" or "sanding rust off an old cabinet." My background reflects this diversity—I've co-authored distributed systems research at UC Berkeley's RISE Lab investigating Byzantine fault tolerance, deployed Stellar validator nodes on Kubernetes, and built smart contracts in Rust and Solidity. I believe this cross-functional technical mindset is exactly what Astra values in maintaining a small, versatile team that punches above its weight.

The mission of improving life on Earth from space resonates deeply with me. I'm eager to contribute to a team where software enables critical engineering decisions, from CAD designs to launch operations, and where my work would have direct, measurable impact on one of the most technically challenging endeavors in aerospace.

I'd welcome the opportunity to discuss how my experience building production systems, optimizing data pipelines, and shipping high-impact solutions can support Astra's mission.

Thank you for your consideration.

Sincerely,

Kentaro Vadney